A Frost-Kill Trading Signal for Black Sea Winter Wheat

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Abstract

The Black Sea region is a central source of world wheat supply. The CME Black Sea Wheat (CVB) contract, active since July 2025, offers direct exposure to it while remaining thinly traded. This study builds an alternative-data supply signal from a Russian-calibrated process-based frost-kill crop model and tests whether it explains wheat futures returns. WOFOST is run with the FROSTOL winter-kill submodule under its MOCCCASIN calibration for Russian winter-wheat genotypes. Frost-kill signals are simulated for highest producing Rostov Oblast, and matched to 21 Euronext milling wheat contracts by harvest year. A binary frost-kill indicator enters a local projection over 30 trading days. The response peaks at +0.72% at horizon 18 but is insignificant at every horizon. Because the coarse reanalysis input, the extrapolated snow model and the proxy contract, the result is unreliable, making it an unsuccessful test on market efficiency.

KEYWORDS: winter wheat, frost kill, crop simulation model, supply fundamentals, wheat futures, local projections

1 Introduction

Despite the ongoing war in Ukraine, the Black Sea region remains one of the most important agricultural production areas in the world. In 2021 the Russian Federation and Ukraine ranked among the top three global exporters of wheat, maize, sunflower seed and sunflower oil. Together they supplied close to 30% of world wheat exports and roughly 80% of world sunflower exports (Food and Agriculture Organization of the United Nations, 2022). Grain and vegetable-oil markets, in particular those for wheat, corn and sunflower, are therefore heavily dependent on the region. That exportable supply is concentrated in only a handful of countries, which leaves it exposed to supply shocks and price volatility.

Reflecting this importance, the newly active Black Sea Wheat contract (CVB, Argus), listed since July 2025, has given hedgers and speculators in commodity derivatives markets a more specific way to quantify their exposure to the Black Sea region. That exposure previously had to be proxied through the CBOT SRW wheat contract on exchanges such as the Chicago Mercantile Exchange (CME). Its contract specifications use the price of 12.5%-protein wheat, free on board (FOB), for Romanian and Bulgarian export prices in the ports of Constanta, Varna and Burgas (CVB) (CME Group, 2025).

As with any such contract, its participants fall into two groups. Commodity derivatives markets offer risk-management tools for hedgers, giving them a more constant capital flow and thereby supporting investment and innovation. Speculators take the other side of these trades, assuming risk with the goal of making a profit.

However, newly initiated futures markets often inherit other risks. Market-depth risk is a sudden price change caused by an order imbalance in a thin market. It matters most for systematic hedgers, and it arises when low trading volume and liquidity leave the order book unable to absorb a trade. The factors that determine market depth, and liquidity more generally, include the rate of transactions during the trading period, the ratio of speculator and scalper activity to overall trading, equilibrium price variability, the size of the market order, expiration-month effects and market structure. Measures such as position limits and warning quotas have been developed to contain it, with the ratio of hedging to speculative volume a recurring element (Pennings & Meulenberg, 1997; Pennings et al., 1998). In the present case the older Platts contract was suspended and replaced by the CVB contract, which has shown low trading volumes and liquidity thus far. The scale of that thinness is visible in the position-reporting data. The CFTC publishes trader positions only for markets in which at least 20 traders hold reportable positions. Through 2026 the CVB contract is absent from every weekly Commitments of Traders report, therefore showcasing that it sits below that reporting floor (Commodity Futures Trading Commission, 2026). This market depth thinness suggests a high ratio of hedgers to speculators among the contract's participants.

Beyond these market-level frictions, the decision to initiate a futures position is not driven by price volatility alone. A manager's risk attitude does not change over the few minutes in which a position is often taken, so risk aversion cannot be the only factor at work. The futures price level at the moment of decision also plays a significant role. It is evaluated against a psychological reference price, defined as a price scale held in the manager's memory against which the current futures price is judged (Pennings, 2002). This behavioural component of price formation in derivatives markets is one that must be accounted for when analysing them.

These factors describe how prices form. The premise of the strategy developed here is that they do not form perfectly. Grossman and Stiglitz (1980) show that a perfectly

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informationally efficient market is impossible. Acquiring and processing information is costly, so prices cannot fully reflect everything that is known. If they did, those who paid to become informed would earn no return on it and would stop paying. In equilibrium some participants are informed and others are not. The informed bear the cost of generating a signal, and they are compensated through the price movements their own trading produces. Those movements reveal their information to the rest of the market only with a lag and with noise. It is this gap, between the moment a supply-relevant signal is generated and the moment it is impounded in the price, that a daily crop-growth frost-kill model may exploit.

The clearest empirical instance of such a costly, weather-based signal is the study of the frozen orange-juice market by Roll (1984). The puzzle there was not that weather failed to matter, but that it appeared to explain almost nothing. Regressing FCOJ returns on temperature surprises produced statistically significant coefficients yet an R^2 of only one to four percent. That result was later taken up by the behavioural-finance literature as evidence that prices do not reflect supply fundamentals. Boudoukh et al. (2007) show that it is a specification artefact rather than a sign of inefficiency, which reinforces the original reading of the market as efficient. Their argument rests on two corrections to the linear, all-season regression. The first is that the relationship is state-dependent. A temperature surprise carries information only when it shifts the market's perceived probability and severity of a freeze, so an unexpected drop from 80 to 50 °F is irrelevant to Florida supply while a far smaller drop from 30 to 25 °F is devastating. Non-freezing observations therefore contain no fundamental signal and must be conditioned out, and the impact is further path-dependent on whether a freeze has already struck that season. The second correction is that the relationship is nonlinear, since damage accelerates convexly as temperature falls below the critical threshold. Once both are imposed, the R^2 rises from roughly 1.5% in the linear model to about 50% on days with freezing temperatures. Although such days are under 4.5% of the winter sample, they account for around two-thirds of winter return variability.

The signal developed here extends this weather-as-information logic into a modern setting. It takes the form of alternative data, meaning information that originates outside the standard repertoire of market data, namely order books, exchange price feeds and scheduled fundamental releases, but that is used to anticipate price moves and shifts in supply or demand (Hansen & Borch, 2022). Such sources tend to come from outside financial markets, to carry a shorter history and to be harder to use than conventional data. Hansen and Borch (2022) describe their adoption as prospecting, the work of rendering a non-standard source usable as a tradable signal, followed by assetization, by which sources once of little interest to trading firms become valuable in their own right. Their key observation is that such sources can act as upstream fundamentals. They capture the state of a supply fundamental before it appears in official production or yield reports, and in turn in price, which gives an informed participant a window to position ahead of the wider market.

One such alternative-data source for CVB wheat futures is WOFOST. WOrld FOod STudies is a daily process-based crop growth model that uses biophysical factors to simulate a crop's end-of-season yield, taking into account weather, soil hydraulic properties and crop-specific physiological parameters (de Wit et al., 2019). Because the target region carries a pronounced risk of frost damage and winter-kill when cultivating winter wheat, an add-on to WOFOST is used, namely FROSTOL. FROSTOL has been parameterised for Russian winter-wheat genotypes with respect to their hardening curve (de Wit, 2013b).

This study treats this alternative frost-kill signal as a upstream supply fundamental to explain CVB-wheat returns. The geographic concentration of deliverable supply in that contract recreates the natural-experiment conditions that made the orange-juice study of Roll (1984) tractable, together with Boudoukh et al. (2007) state-dependency into account by looking only at agronomically-sensible frost-risk periods.

2 Data & Methods

2.1 Model

The supply signal is generated with WOFOST (de Wit et al., 2019), run with the FROSTOL winter-kill submodule (Bergjord et al., 2008). It is implemented in the open-source Python Crop Simulation Environment (PCSE). FROSTOL originates as a Norwegian model (Bergjord et al., 2008) that simulates the course of frost tolerance. That tolerance is expressed as the state variable LT_{50}, and it is driven by crown temperature, snow depth and vernalisation stage.

FROSTOL represents the frost tolerance of a winter wheat stand through a single state variable, LT_{50}, defined as the temperature at which half of the plants die (Bergjord et al., 2008). The state is initialised at sowing and integrated forward on a daily step,

$$\mathrm{LT}_{50,t} = \mathrm{LT}_{50,t-1} - R_h + R_d + R_r + R_{ts}. \tag{1}$$

Here R_h is the rate of hardening. The three remaining terms are rates of dehardening, driven by rising temperature, by respiration stress and by prolonged exposure to near lethal temperature. All four rates carry units of degrees Celsius per day. Two variables drive the model, snow depth and the mean daily crown temperature T_c, which is the temperature experienced at the wheat crown a few centimetres below the soil surface.

Hardening depends on the distance between the current state and the critical LT50 (LT_{50c}), the maximum tolerance a cultivar can attain under full hardening,

$$R_h = \begin{cases} P_h \left(10 - T_c^{\lim}\right) \left(\mathrm{LT}_{50,t-1} - \mathrm{LT}_{50c}\right), & T_c < 10 \\ 0, & T_c \geq 10, \end{cases} \tag{2}$$

where $T_c^{\lim}$ is the crown temperature clipped to the interval $[0, 10]$ and P_h is the hardening parameter. Hardening proceeds only while the vernalisation requirement remains unmet. Once vernalisation is satisfied R_h is set to zero and the crop can no longer gain tolerance.

The first dehardening term captures the loss of tolerance as temperature rises,

$$R_d = \begin{cases} 0, & T_c < T_{\text{crit}} \\ P_d \left(\text{LT}_{50,i} - \text{LT}_{50,t-1}\right) (T_c + 4)^3, & T_c \geq T_{\text{crit}}, \end{cases} \tag{3}$$

with $\text{LT}_{50,i}$ the initial tolerance at sowing, P_d the dehardening parameter and T_{crit} the crown temperature at which dehardening begins. T_{crit} equals 10 °C before vernalisation and falls to −4 °C afterwards, so a vernalised crop dehardens rapidly as temperature rises.

The second term represents respiration stress, which acts when soil temperature is near zero under a snow layer,

$$R_r = P_r * \text{RE} * \text{RSD}, \qquad \text{RE} = \frac{\exp(0.84 + 0.051\,T_c) - 2}{1.85}, \tag{4}$$

where P_r is the respiration parameter and RSD is relative snow depth, rising linearly with snow depth to 12.5 cm and constant beyond that point.

The third term reflects the experimental finding that survival declines under sustained exposure to temperatures close to the lethal threshold, even where that threshold is never crossed,

$$R_{ts} = \frac{\text{LT}_{50,t-1} - T_c}{-P_{ts}\left(\text{LT}_{50,t-1} - T_c\right) - 3.74}, \tag{5}$$

with P_{ts} the temperature stress parameter. The hardening, dehardening, low-temperature and respiration-stress rate coefficients are held at the FROSTOL defaults of Bergjord et al. (2008), which that calibration retains because they are largely cultivar-independent.

Within the *"Monitoring Crops in Continental Climates through Assimilation of Satellite Information"* (MOCCCASIN) project (de Wit, 2013a, 2013b), FROSTOL was parameterised for Russian winter-wheat genotypes. Its single region-specific parameter, the critical LT_{50}C, was set to −18 °C from the documented gradient of cold hardiness across Russian cultivars (Fedoulov, 1998), and it was embedded in an operational winter-wheat monitoring system for the Tula region of European Russia.

FROSTOL returns a tolerance state rather than a damage estimate, so a further step is required to translate that state into plant loss. Within MOCCCASIN this is done with a logistic kill function,

$$K_t = 100\% \times \left[1 + \exp\left(\frac{T_c^{\min} - \text{LT}_{50,t}}{P_{kcf}}\right)\right]^{-1}, \tag{6}$$

where K_t is the percentage of plants killed on day t and $T_c^{\min}$ is the minimum daily crown temperature, used in place of the daily mean because damage follows the coldest point of the day. The form returns 50% whenever the minimum crown temperature equals the current state, and the steepness parameter P_{kcf} (KILLCF) is set to 1.019 (de Wit, 2013a). The logistic kill function that converts the LT_{50} state into a percentage of plants killed explained 96.9% of the variance in observed mortality from controlled freezer tests of winter wheat.

Within MOCCCASIN, the model reproduced the observed spatial pattern of the severe 2010 winter-kill across the region, with the strongest simulated impact near Uzlovaya, Volovo and Suvorov, matching the stations that recorded the worst damage. This agronomic validation for the northern-region is crucial, but not market-relevant since has little influence on region-wide supply.

By leaving vernalisation enabled (IDSL ≥ 2, as FROSTOL requires), the frost-tolerance state LT_{50} therefore evolves continuously across the full hardening-and-dehardening cycle, so that both the midwinter and the late-spring kill regimes are represented. Damage in FROSTOL is driven by the crown temperature rather than the air temperature, so the submodule is coupled to a CrownTemperature model that maps daily air temperature to crown temperature as a function of snow depth. This coupling is central rather than incidental. Because snow is a poor conductor of heat it insulates the crown, so that under full snow cover the minimum crown temperature stays above roughly −8 °C even when the air falls to −35 °C, whereas over bare ground the crown closely follows the air. FROSTOL therefore requires a daily snow depth, which the Russian station network reports only once every ten days. Following MOCCCASIN, this is supplied by SnowMAUS (Trnka et al., 2010), which simulates snow accumulation and melt as a snow-water-equivalent and converts it to a depth, using the Tula Oblast snowdensity as calibration. Hereby, the Tula-calibrated SnowMAUS model was capable of replicating the snowdynamics across snow-depth observations of meteorological stations in the Tula region, despite uncertainty on snow density. Further, the standard WOFOST winter-wheat parameterisation set (Winter_wheat_101) from the WOFOST crop-parameter repository has been used (de Wit, 2020).

The model is run only in cells that carry a meaningful amount of winter wheat, since those are the cells whose production actually affects the price. Per-cell wheat area comes from CROPGRIDS (Tang et al., 2024), a global gridded crop-area product, summed onto the 0.5° NASA POWER grid. A cell is kept when wheat covers at least 10% of its area, rather than when it passes a fixed number of hectares, so the same standard applies to every cell regardless of its size. Cells with only a trace of wheat are dropped, because they carry too little production to move the price the signal is built around. Hereby, the mask sets where the model runs.

One limitation of this weight is that CROPGRIDS reports a single all-wheat figure and does not separate winter from spring wheat. A cleaner version would combine the CROPGRIDS wheat area with a winter-specific layer such as the NASA GLAM winter-grains product for Russia. This limitation is not so crucial since this study

Table 1. Winter-kill and snow parameters. FROSTOL uses the MOCCCASIN Tula calibration for Russian winter wheat and SnowMAUS supplies simulated snow. The crop uses the standard WOFOST `Winter_wheat_101` parameterisation with vernalisation enabled (IDSL ≥ 2) under `Wofost_winterkill.conf`.

Parameter	Meaning	Value	Unit	Source[a]
FROSTOL winter-kill (MOCCCASIN Tula calibration)				
`LT50C`	Lowest attainable LT_{50} (max. frost tolerance)	−18.0	°C	F98, D5.3
`LT50I`	Initial LT_{50} at sowing	−4.01	°C	D5.3
`FROSTOL_H`	Hardening rate coefficient	0.0093	–	B08, D5.3
`FROSTOL_D`	Dehardening rate coefficient	2.7×10^{-5}	–	B08, D5.3
`FROSTOL_S`	Low-temperature stress coefficient	1.910	–	B08, D5.3
`FROSTOL_R`	Respiration (snow-cover) stress coefficient	0.541	–	B08, D5.3
`FROSTOL_KILLCF`	Logistic kill-function steepness	1.0189	–	dW, D5.3
`FROSTOL_SDBASE`	Snow depth at which respiration stress begins	0.0	cm	D5.3
`FROSTOL_SDMAX`	Snow depth at which respiration stress saturates	12.5	cm	D5.3
`CROWNTMPA`	Crown-temperature coefficient *A*	0.2	–	D5.3
`CROWNTMPB`	Crown-temperature coefficient *B*	0.5	–	D5.3
SnowMAUS snow model				
`TMINACCU1`	T_{min} above which precipitation stays rain	0.0	°C	T10
`TMINACCU2`	T_{min} below which precipitation falls as snow	−6.0	°C	T10
`TMINCRIT`	T_{min} below which the snowpack does not melt	−12.0	°C	T10
`TMAXCRIT`	T_{max} above which melt occurs	5.0	°C	T10
`RMELT`	Degree-day melt rate above the melt threshold	0.042	cm/(°C d)	T10
`SCTHRESHOLD`	Snow-water-equivalent threshold for snow cover	2.0	cm	T10
`SNOWDENSITY`	Snow density (SWE to snow depth)	0.164	–	D5.3

[a] F98 = Fedoulov (1998), B08 = Bergjord et al. (2008), T10 = Trnka et al. (2010), D5.3 = MOCCCASIN Deliverable D5.3 (de Wit, 2013b). FROSTOL rate coefficients are left at their Bergjord et al. (2008) defaults, retained by D5.3.

solely on the signal of the Rostov Oblast. As the only Russian-calibrated FROSTOL parameter $LT_{50}C$ (−18 °C) was derived from Don-region cultivars (Fedoulov, 1998), together with the fact this southern oblast is not only Russia's but also the Black-Sea's top producing region, thus producing the most reliable & influential signal for market-wide winter-wheat disruptions.

2.2 Signal Workings

The mechanism that makes this state tradable is the same nonlinear, state-dependent structure that Boudoukh et al. (2007) impose on orange-juice returns, except that here it follows from crop physiology rather than from fitting returns. FROSTOL represents frost tolerance as the single state variable LT_{50}, the temperature at which half of the plants die. This variable is updated daily from sowing, so it falls as the crop hardens through winter and rises again as it dehardens towards spring. Its driver is not the air temperature but the crown temperature, the temperature at the wheat crown a few centimetres below the soil surface, which snow insulates from the air above. Damage then depends on the gap between the minimum crown temperature and the current LT_{50}, through a logistic kill function, so the same minimum temperature is harmless to a fully hardened crop and lethal to a dehardened one. The danger is greatest late in the season (Spring), where two defences fall away at once. The snow pack has melted, so the crown is exposed and again tracks the air, and the LT_{50} has risen as the crop dehardens after vernalisation, so its tolerance is shallow. A moderate spring freeze on an exposed, dehardened crop can therefore kill more than a deeper midwinter freeze that a hardened, snow-covered crop survives, which is exactly why late Spring frosts are treated as the most dangerous (see Figure 1).

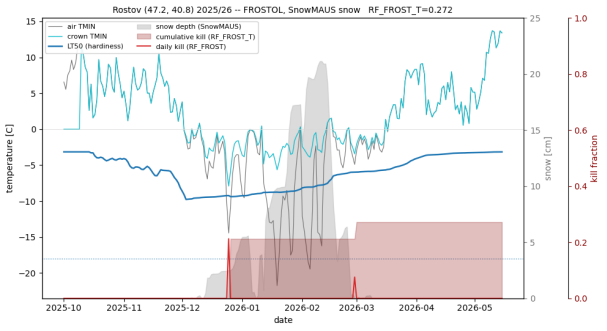


Figure 1. Rostov cell (47.2, 40.8). FROSTOL within-season diagnostics for 2025/26 winter-wheat season. The top row shows air and crown minimum temperature against the dynamic LT_{50}, with the dotted line marking the $LT_{50}C$ hardiness floor (−18 °C). The lower rows show simulated snow depth, daily kill, and cumulative kill fraction RF_FROST_T.

FROSTOL also represents the carbohydrate cost of prolonged snow cover through a respiration-stress term that accelerates dehardening and contributes to kill in proportion to the depth and persistence of the snowpack (Bergjord et al., 2008). A crop under deep, lasting snow

stays metabolically active, depletes its reserves and loses hardiness, so a deep-snow winter can register damage even without a severe frost. The signal therefore separates two regimes that a fixed temperature threshold would blur together. One is cold kill of an inadequately hardened crop. The other is reserve depletion under prolonged snow cover.

The market observes temperature, which is public and priced almost at once, but not the hardening state, which the model supplies and which decides whether a given cold spell is harmless or lethal. A frost scare therefore tends to move the price by roughly the same amount whatever the true damage, because the market reacts to the visible cold rather than to the hidden state. That move then reverses as the actual impact is reassessed over the following weeks (M. Tannura, personal communication, 2026).

Two boundaries of the analysis follow from this framing. First, because price formation is the meeting of supply and demand, and the strategy is built only on supply-side factors, demand is assumed constant in the short term. Second, the signal quantifies the quantity risk carried in these futures markets. The price risk, meaning the hedging and basis risk that a futures position introduces even as it reduces spot-price risk, remains unexplained by the signal and is carried as a residual exposure of the strategy.

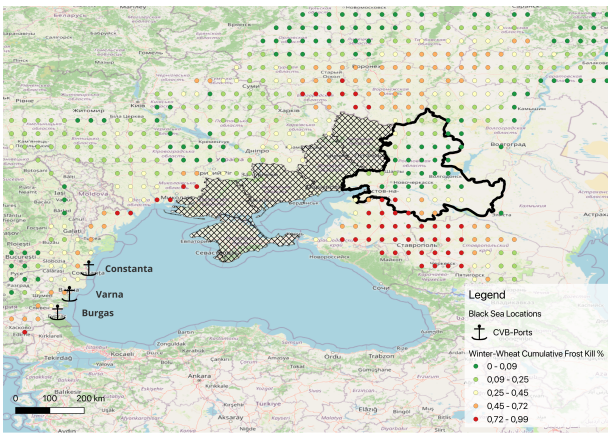


Figure 2. Cumulative season kill fraction (RF_FROST_T) across the Black Sea winter-wheat belt (2025/26). Bulgaria and Romania are uncalibrated extrapolation, shown for context only. The defensible signal is the Russian Rostov Oblast (see black border).

2.3 Matif

The Euronext Milling Wheat (MATIF) futures contract is used as the price series in place of the CME Black Sea Wheat (CVB) contract. Although the two contracts are not identical in their delivery and quality specifications, they represent closely related classes of internationally traded milling wheat and are consequently exposed to a common set of global and regional wheat-market fundamentals. The use of a related futures contract as a proxy when the preferred contract is insufficiently liquid is consistent with the cross-hedging literature, which establishes that a futures contract need not have an identical underlying to provide information about, or exposure to, the price risk of a closely related commodity (Anderson & Danthine, 1981). This consideration is particularly important for newly established futures markets, for which limited trader participation and liquidity can impede the informational quality and empirical usefulness of observed prices (Cuny, 1993). In wheat markets specifically, Janzen and Adjemian (2017) show that futures contracts representing different wheat classes and geographical markets nevertheless contribute to the discovery of a common world wheat price, including the Paris wheat futures market. MATIF therefore provides a substantially longer and more liquid price history while retaining exposure to the wheat-market fundamentals relevant for the present analysis. The validity of this substitution is also supported directly in the available sample: daily returns on the MATIF and CVB contracts have a correlation of $r = 0.73$ across 299 overlapping observations after converting MATIF prices into U.S. dollars (Fig. 3). MATIF is consequently treated not as an identical substitute for CVB, but as a liquid empirical proxy for the wheat-price response that the considerably thinner CVB series is intended to represent.

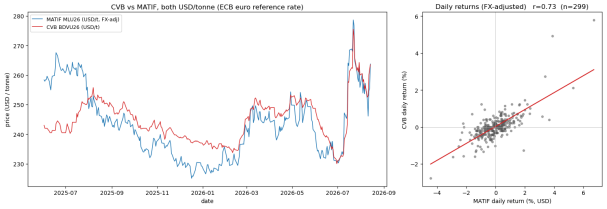


Figure 3. Correlation between CME Black-Sea Futures contract due September 2026 (BDVU26, red line), and MATIF MLU26 adjusted to USD using the ECB euro reference rate (blue line). Return correlation of $r = 0.73$ across $n = 299$.

2.4 Local Projections

Local projections, established by Jordà (2005), are used to examine how wheat prices absorb the information contained in a frost-related supply shock over time. This approach follows a broader empirical finance literature in which local projections are used to distinguish the initial response of an asset price from its subsequent adjustment as information is incorporated by the market. Most closely related, Accominotti et al. (2024) use local projections to trace the short-run adjustment of the same sovereign bonds traded across different markets following a price innovation in one market, allowing them to measure the speed with which information is transmitted and arbitrage restores price alignment. Similarly, Avramov et al. (2026) use local projections to follow stock returns after innovations in economically related firms, interpreting the subsequent path of returns as evidence on whether information is incorporated immediately or diffuses only gradually into prices. More broadly, Jordà et al. (2015) employ local projections to trace the dynamic response of house prices to identified monetary shocks, illustrating their usefulness

for studying the persistence of asset-price responses rather than only their contemporaneous effect. This distinction is central to the present analysis. A frost event represents new information about prospective wheat supply, but the relevant economic question is not restricted to the price movement on the day on which that information becomes available: prices may adjust gradually as the extent of crop damage becomes clearer, or an initial reaction may subsequently reverse if the perceived supply loss is not confirmed. Local projections allow these alternative adjustment paths to be observed directly across successive horizons.

3 Results

As mentioned before, the MOCCCASIN calibration is done based on Don-based winter-wheat genotypes, hereby making the Rostov Oblast the calibrated region for the critical $LT_{50}C$ of −18 °C (de Wit, 2013b; Fedoulov, 1998). This is very useful since the Rostov Oblast is the highest winter wheat producing region in Russia (Abys et al., 2022) and one of the highest winter-wheat cropland area holding 17.2% (Abys et al., 2024), hereby making it the suited parameterized region that should facilitate meaningfull supply shocks in case frost-kill events happen.

For each wheat pixel in the Rostov Oblast the daily frost-kill is simulated. Further, each grid cell is then weighted by its wheat area (Tang et al., 2024). By multiplying each cell's kill by its weight, and then adding them up, we come out to a single Rostov signal for that day.

Following McKenzie, 2008, who uses new-crop December corn and November soybean futures to study pre-harvest USDA report surprises, this seasonal winter-wheat frost kill supply signal is matched to the first Euronext MATIF milling wheat contract delivering the affected harvest. Winter-wheat is harvested in early summer, hereby the contract is November for harvest years 2006 to 2014 (MLX06 to MLX14) and September for harvest years 2015 to 2026 (MLU15 to MLU26). This reflects the change in contract cycle from November, January, March and May until May 2015, then September, December, March and May from September 2015 onwards (Euronext, 2018).

Hereby, the bio-physical crop model simulates 21 winter-wheat seasons, whereas each of the 21 Euronext Milling Wheat contracts is matched to the winter-wheat harvest it delivers.

Further, for analysis simplification, the shock has been made binary such that days with frost-kill were assigned as 1, otherwise 0 (D_t). Doing so, the severity of the frost-kill that FROSTOL simulates is not taken into account as signal.

From the 4770 of winter-wheat simulation days, only 113 frost-kill event days were captured in 18 of the 21 winters. By using the local projection method, we want to see how MATIF responds to a frost-kill signal over a time horizon of 30 trading days. For each horizon h, the cumulative return over the h trading days following a frost-kill event are compared to the cumulative return over the same h trading days following a day on which the signal did not happen.

This cumulative return is measured from the settlement price of the day before the signal, so that for every trading day in the sample

$$r_{t,t+h} = 100\left(\ln P_{t+h} - \ln P_{t-1}\right), \tag{7}$$

, which means the price movement on the day the signal arrives is part of the measured response. For example, the mean cumulative return after frost-kill days is +1.06% (n = 113), whilst on non-frost days the return equals +0.27% (see Figure 4). The raw difference thus between a day on which the frost signal hits and not such a day is +0.80%.

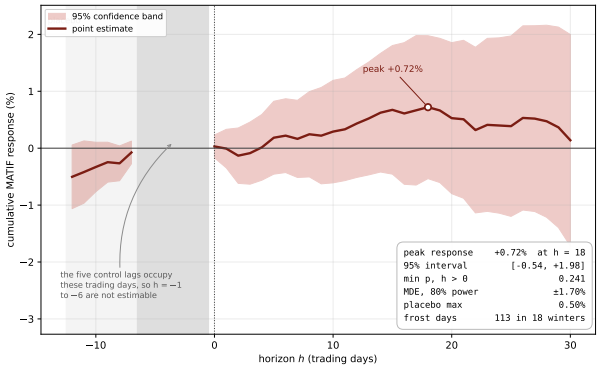


Figure 4. Local Projection with horizon of 30 trading days in the future.

To control and uncredit unjustified price rallies before a frost signal, the projection includes five preceding returns (1 whole trading week), hereby meaning that the frost response is measured against days that entered with a comparable recent price movement, rather than against all other days regardless of what the market had just done (h = [-1, -5]).

Written out, for each horizon

$$r_{t,t+h} = \alpha_h + \beta_h D_t + \sum_{k=1}^{5} \gamma_{h,k}\, r_{t-k} + \varepsilon_{t,h}, \qquad h = -12, \ldots, 30, \tag{8}$$

, where D_t is the binary frost-kill indicator and the summation term collects the five preceding daily returns, which enter as controls rather than as objects of interest. The coefficient β_h is the difference the projection measures at horizon h. Because the outcome windows of consecutive days overlap almost completely, the residuals are serially correlated by construction, and standard errors are computed using the Newey and West (1987) correction. Hereby, the controlled difference 18 trading days after a frost signal happened is +0.72% (see Figure 4).

Across all 30 forward horizons, the smallest p-value is 0.241, so no horizon approaches conventional significance and the response never separates from zero. At the peak horizon the 95% interval runs from −0.54 to +1.98%.

Horizons before the event act as a falsification test, since frost that has not yet happened cannot move prices. The largest pre-event coefficient is −0.50% against a peak of +0.72% after. The post-event response is therefore no

larger in absolute terms than what the design produces where no response can exist.

Alongside the local projection, the peak-kill day of each crop year is taken as a single event, so that every year contributes exactly one observation and no year dominates the comparison simply because its frost signal happened to fire on more consecutive days than another. For each of these events the cumulative return is traced from ten trading days before the peak-kill day to sixty trading days after it, measured relative to the day before the event so that every path begins at zero on the event day (see Figure 5).

Panel (a) plots those paths together, with the median in black and the interquartile range shaded, and it shows that the spread between crop years is far larger than any central tendency, with the strongest year reaching roughly +23% and the weakest around −16% while the median remains close to two per cent throughout. Panel (b) presents the same paths as one row per crop year, sorted so that the most severe simulated frost sits at the top, which means that if severity drove the price response the red would concentrate in the upper rows. It does not, hereby showcasing that signal should be better read as binary.

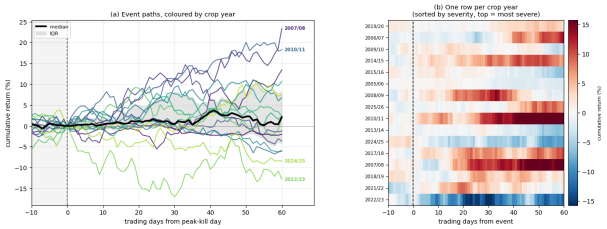


Figure 5. Event Study, where panel a) shows the each crop season's major frost-kill event's returns after 60 trading-days, and panel b) the crop year sorted by frost-kill severity, compared to return.

4 Discussion & Limitations

Daily reanalysis weather input from global gridded meteorological datasets such as NASA POWER is first of all an uncertain and smoothened representation of past weather events, and this matters a great deal for a study basing itself on the explanatory power of a daily signal in an end-of-day market. Hogan and Schlenker (2024) warn that coarse daily weather data may carry substantial measurement error causing attenuation bias, and that this is especially important for studies of non-linear effects, because spatial averaging misplaces temperature exposure across the critical threshold separating beneficial from harmful temperatures. They find that about 55% of the predictive gap between fine-scaled and coarse data is due to spatial resolution alone, with the remainder attributed to the input observations and to the reanalysis procedure, which may mask the most extreme weather events.

Their own results temper this warning and it should be reported. All three weather data sets in their comparison estimate similar response functions, and the difference between fine and coarse products appears in out-of-sample predictive skill rather than in the estimated coefficients. Coarsening the record in time, from hourly values to a daily minimum and maximum, has almost no effect at all.

What matters here is not the coefficient but the date on which the signal fires. FROSTOL was coupled into WOFOST through a spring field plant coverage factor with the aim of obtaining more reliable estimates of winter wheat yields for regions in Russia, as reported by Bergjord Olsen et al. (2018). For that purpose only the total kill accumulated across the winter needs to be correct, and it does not matter whether an increment is assigned to the fifth or the seventh of February. This study needs the opposite, because the return window opens on the day the event fires, so any smoothing of the weather input that moves an increment onto a neighbouring day moves the entire event window with it.

The model currently runs with the SnowMAUS add-on, which was calibrated to the northern Tula Oblast's snow density observations. Even though this snow depth model captured the Tula region's seasonal snow dynamics well, extrapolating it to the more southern Rostov region is unreliable, and the consequence of a snow depth error is not marginal. Bergjord Olsen et al. (2018) report that raising snow depth at the Swedish location Motala from 0 to 5 cm between 30 January and 10 February would increase the number of cultivars classified as surviving from two to five and raise model accuracy from 40% to 80%. If local meteorological stations recording daily snow depth were accessible, reliable and suitable for model input, this source of error would be removed rather than bounded.

On the other hand, given previous Russian cultivar specific frost tolerance work, the extrapolation of the frost tolerance parameters to Rostov Oblast is appropriate, since these were taken from Don-region specific winter wheat genotypes.

CROPGRIDS is a further simplification, since it gives the harvested area for wheat as a whole rather than for winter wheat specifically. In Rostov Oblast the two are not identical, so the production weights applied across cells are approximate. This could be improved by using a winter wheat specific mask such as the one distributed with NASA GLAM.

The same extrapolation problem appears on the price side. MATIF is treated above not as an identical substitute for CVB but as a liquid empirical proxy, and the correlation of $r = 0.73$ that supports the substitution is measured on returns as a whole rather than on the response to Black Sea supply news specifically. Much of that co-movement plausibly reflects the common world wheat price that Janzen and Adjemian (2017) document, so it does not establish that a regional supply shock passes into the two contracts to the same degree. A shock originating in Rostov Oblast may be priced more directly in CVB, where it is central to the delivery specification, than in MATIF, where it competes with European and global fundamentals. If so, the coefficient estimated on MATIF is a scaled-down version of the one that would be estimated on CVB, and the correlation alone does not identify the

size of that scaling. Even still, the signal sits within the most productive region, Rostov, while the CVB contract settles against Romanian and Bulgarian port prices, so a position in the CVB market of presumes that a supply shock in the Russian belt passes through the Black Sea free-on-board complex to that delivery basis.

All in all, the link between fundamental supply information and market pricing has been difficult to prove empirically before. Karali et al., 2020 have studied this using USDA Crop Production report expectation reports for corn, soybean and wheat futures markets. Using traditional OLS-method only 27% was explainable, but after correcting for measurement errors, the same fundamental supply information explains 78% of variance in return with the ICT method. This method however can only be applied to information publicly available to & priced in by all market participants, unlike the crop-growth frost-kill model signal used here.

Further, this signal insignificance may also be linked to geopolitical tensions in recent years the region has been facing. The Black Sea Grain Initiative, signed in Istanbul on 22 July 2022, was not renewed after its third term expired on 17 July 2023 (United Nations, 2023). The export corridor itself has since become a target. Navigation in the Sea of Azov, the outlet for Rostov Oblast's own grain exports, has been suspended since July 2026, and all three grain terminals at Novorossiysk halted operations in August, taking more than 90% of Russian grain export capacity in the Azov-Black Sea basin offline (The Moscow Times, 2026). These are not taken account in the signal analysis, which attributes post-event returns to the frost-kill increment alone while the same region was moving wheat prices through a logistical channel unrelated to the biophysical one modelled here.

The estimates reported here are close to zero and statistically insignificant at all horizons, and the limitations set out above mean this cannot be read as evidence on market efficiency. Insignificance is what one would expect if the market already prices the underlying frost event, and it is equally what one would expect if the signal is measured too coarsely to detect a response that is in fact there. The spatial averaging of the weather input, the snow depth extrapolation and the substitution of MATIF for the CVB contract each pull the estimate toward zero, so the two explanations are not separable within this design. What can be done is to further validate the signal to get reliable results.

5 Conclusion

This study asked whether a frost-kill signal built from a process-based crop model carries information about Black Sea wheat futures returns. Across 113 event days in 18 of 21 winters, the local projection returns a peak response of +0.72% at horizon 18 with a 95% interval of $[-0.54, +1.98]$, and no horizon reaches significance.

Three limitations prevent this from distinguishing an efficient market from an imprecise signal. The reanalysis input is a grid-cell average, so a cold snap may be smoothed onto neighbouring days, which matters because the return window opens on the day the signal fires. The crucial snow depth variable is calibrated to the northern Tula region and applied to Rostov. Further, MATIF proxies a CVB contract that itself settles against Romanian and Bulgarian ports, so two basis gaps separate the modelled region from the priced one. Each pulls the estimate toward zero, which is also where an efficient market would put it.

The contribution is therefore methodological. The paper shows that a calibrated crop growth model can be operated as a daily, location-level supply signal and tested directly against futures returns, and delivers a bound rather than a finding. Separating the market explanation from the measurement explanation requires validating the signal against realised Rostov yield, independently of any price.

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